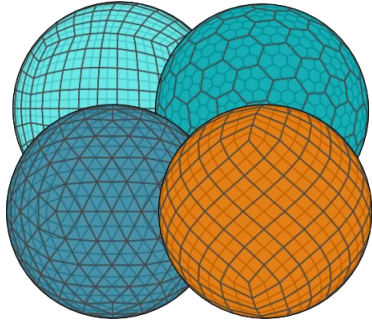
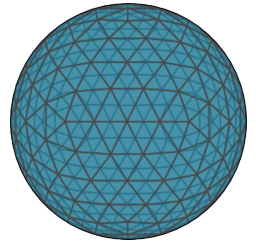




xdggs

Using Discrete Global Grid Systems with **xarray**

Acknowledgements

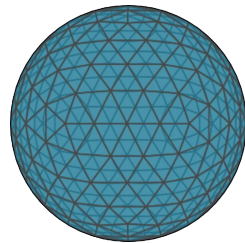


xdggs was initiated using funding from CNES (PANGEO IAOCEA, contract R&T R-S23/DU-0002-025-01) and the European Union (ERC, WaterSmartLand, 101125476, Interreg-BSR, HyTruck, #C031).

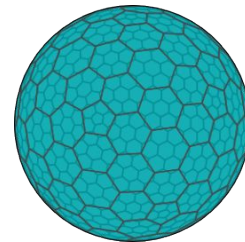


Outline

- Motivation (or: why are traditional map projections not good enough?)
- Introduction to DGGS
- Using DGGS with xarray



Motivation – Data cubes



xarray.Dataset

Dimensions: (time: 39, latitude: 2041, longitude: 4320, elevation: 50)

Coordinates:

elevation	(elevation)	float32	-5.728e+03 -5.275e+...	
latitude	(latitude)	float32	-80.0 -79.92 -79.83	
longitude	(longitude)	float32	-180.0 -179.9 ... 179....	
time	(time)	datetime64[ns]	2021-07-01 ... 2024-...	

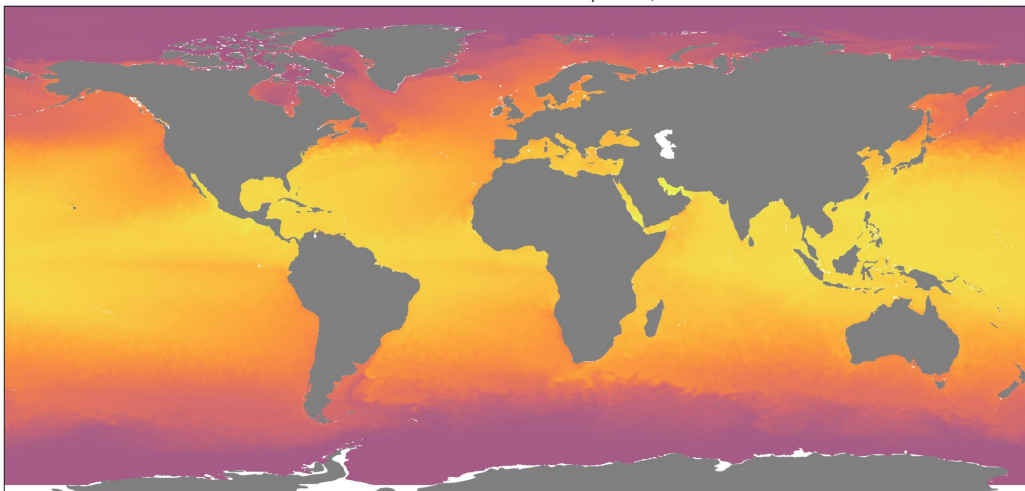
Data variables:

bottomT	(time, latitude, longitude)	float64	dask.array<chunksiz...	
mlotsT	(time, latitude, longitude)	float64	dask.array<chunksiz...	
siconc	(time, latitude, longitude)	float64	dask.array<chunksiz...	
sithick	(time, latitude, longitude)	float64	dask.array<chunksiz...	
so	(time, elevation, latitude, longitude)	float64	dask.array<chunksiz...	
thetao	(time, elevation, latitude, longitude)	float64	dask.array<chunksiz...	
uo	(time, elevation, latitude, longitude)	float64	dask.array<chunksiz...	
usi	(time, latitude, longitude)	float64	dask.array<chunksiz...	
vo	(time, elevation, latitude, longitude)	float64	dask.array<chunksiz...	
vsi	(time, latitude, longitude)	float64	dask.array<chunksiz...	
zos	(time, latitude, longitude)	float64	dask.array<chunksiz...	

Indexes: (4)

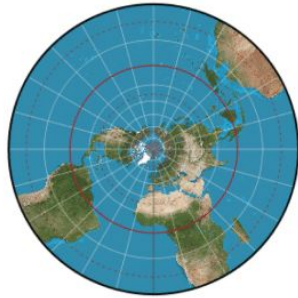
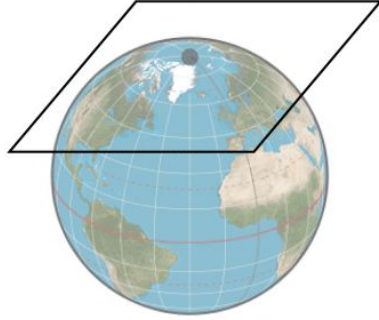
Attributes: (14)

CMEMS GLORYS12V1 ocean surface water temperature, 0.083° × 0.083°

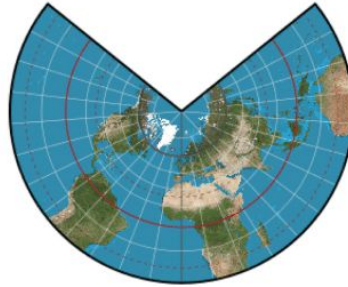
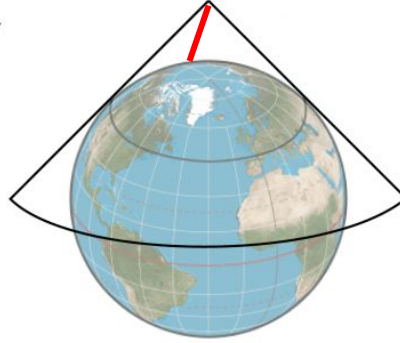


Model: <https://doi.org/10.48670/moi-00021>

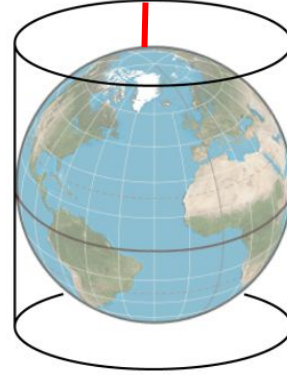
Motivation – Map Projections



Planar



Conal



Cylindrical

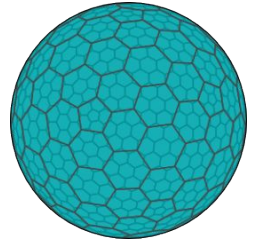
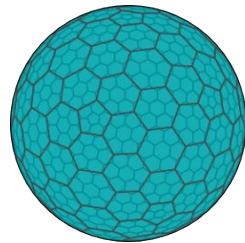


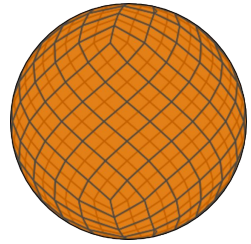
Image from gistbok.ucgis.org

Motivation – Map Projections

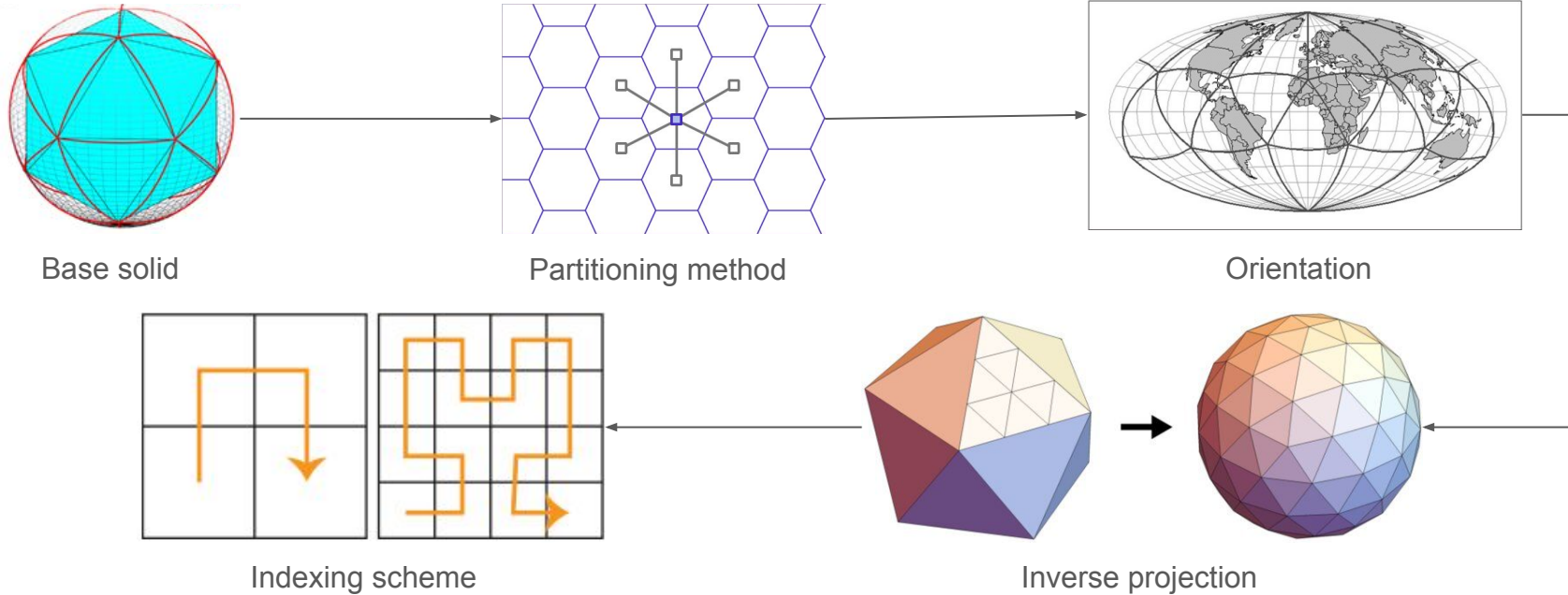


- Advantages:
 - Simplicity
 - Closer to 2D visualization
- Disadvantages:
 - Distortion
 - For datasets with local projections: interaction with other datasets (→ reprojection)
 - Discontinuities

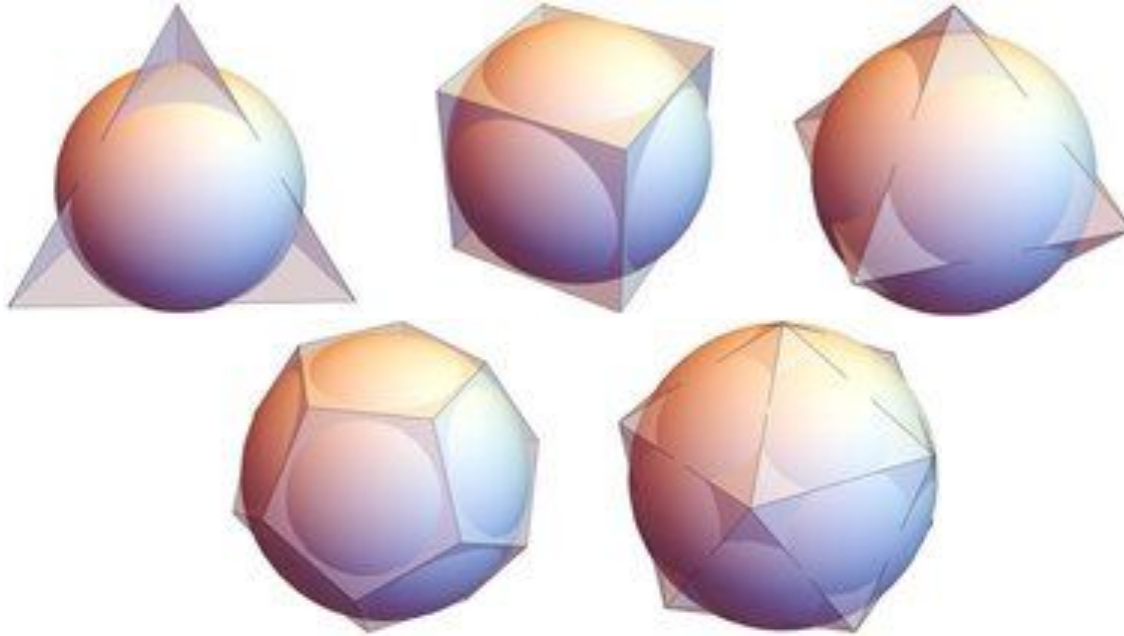
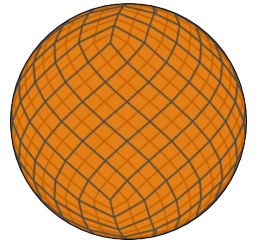
Discrete Global Grid Systems



“A Discrete Global Grid System is a spatial reference system that uses a hierarchical tessellation of cells to partition and address the globe.” (OGC Abstract Specification, 2017).



Initial subdivision: Base solids



- Common base solids:
 1. Tetrahedron
 2. Cube
 3. Octahedron
 4. Dodecahedron
 5. Icosahedron
- Can also use spherical geometry instead
- Choice limits the cell shape
- The smaller the face, the less it is distorted

Orientation

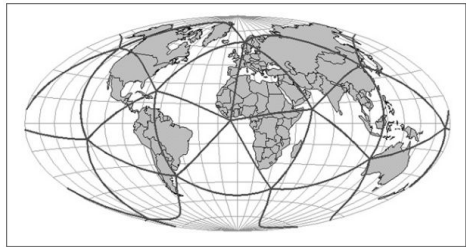
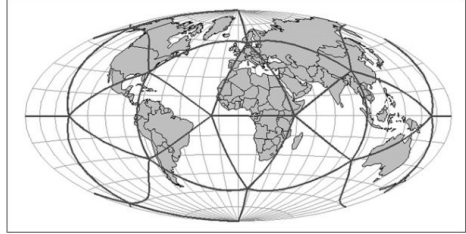
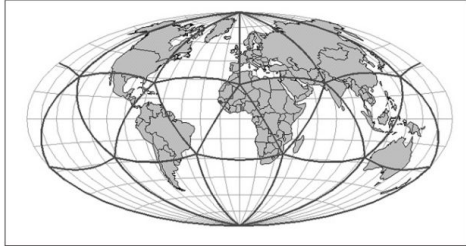
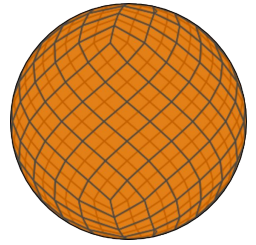


Image from Sahr et al., 2004

- Choice depends on the application
- Aim: keep areas of high distortion away from areas of interest
- Examples for the orientation
 1. Vertices at the poles
 2. Symmetrical to the equator
 3. All vertices in the ocean

Partitioning method

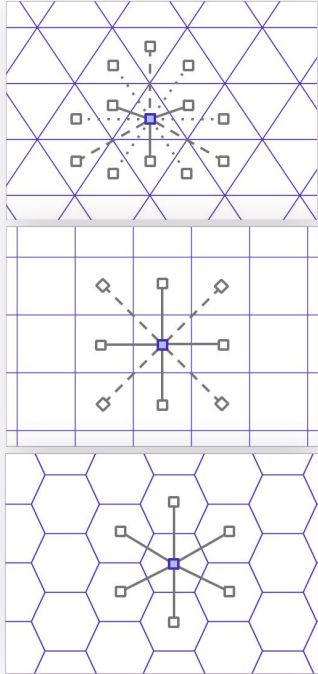
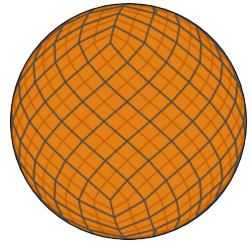


Image from h3geo.org

- Method used to recursively subdivide the faces
- Defines the shape of the cells
- Possible choices:
 - Triangle
 - Square
 - Diamond
 - Hexagon
- Properties:
 - Congruence (possibility of the seamless subdivision of a cell)
 - Aperture (number of parts the cell is subdivided into)
 - Adjacency (relation to neighbours)
 - Compactness (area to perimeter ratio)

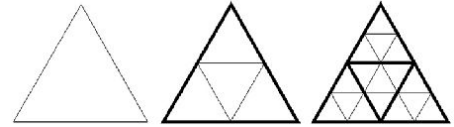
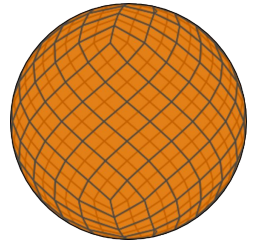


Image from Sahr et al, 2013

Transformation to the spheroid's surface



- Inverse map projection
 - Trade-off between area and shape distortion
 - Area preservation: Snyder equal area
 - Shape preservation: Gnomonic
 - Compromises: Fuller - Gray

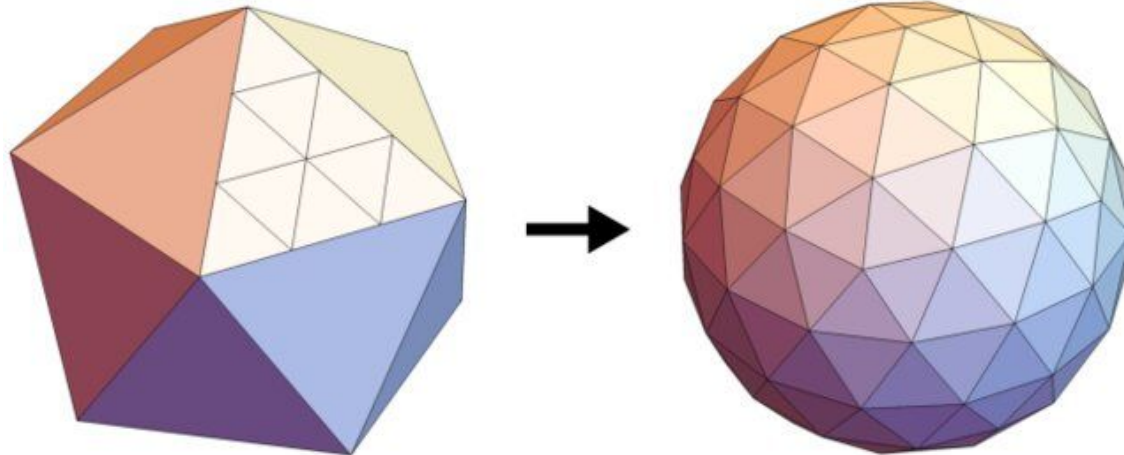
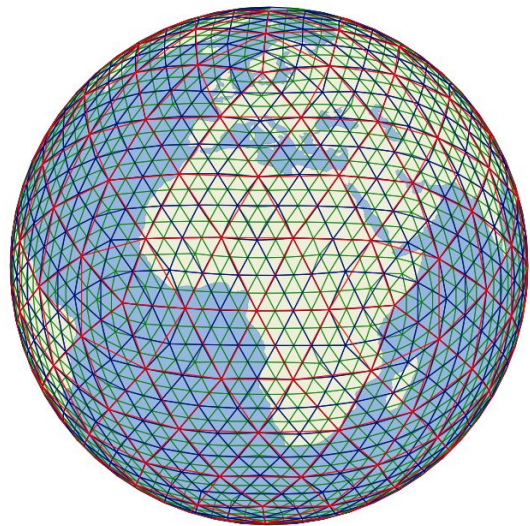
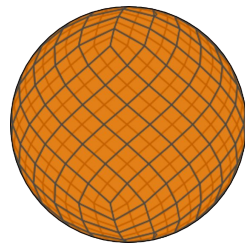
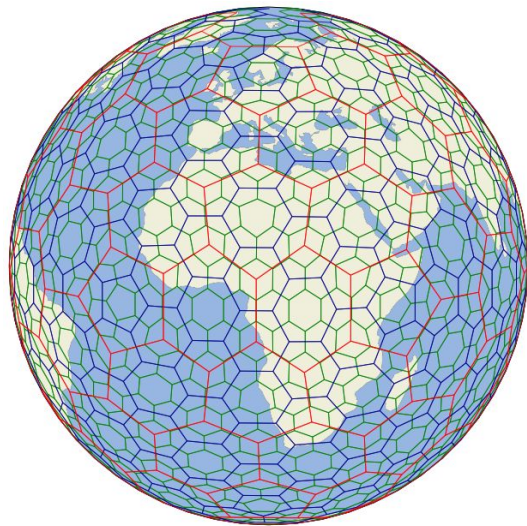


Image from: Brower et al, 2020

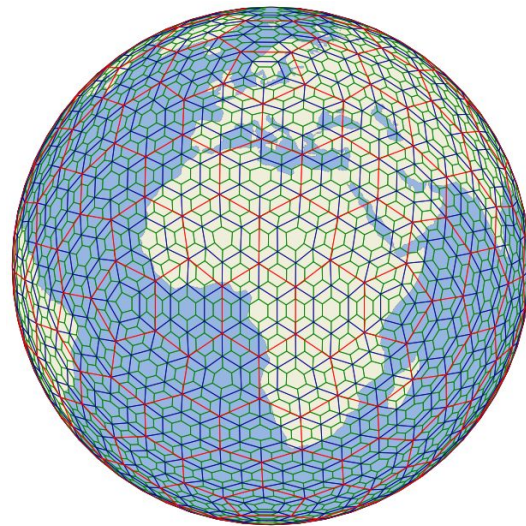
Examples for DGGS



ISEA4T

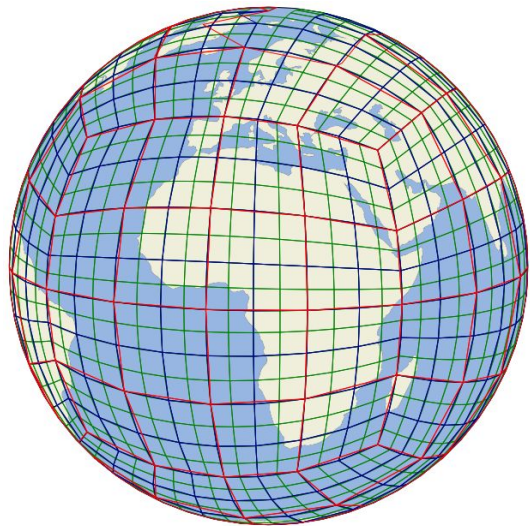
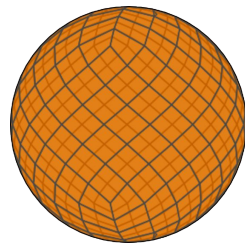


ISEA3H

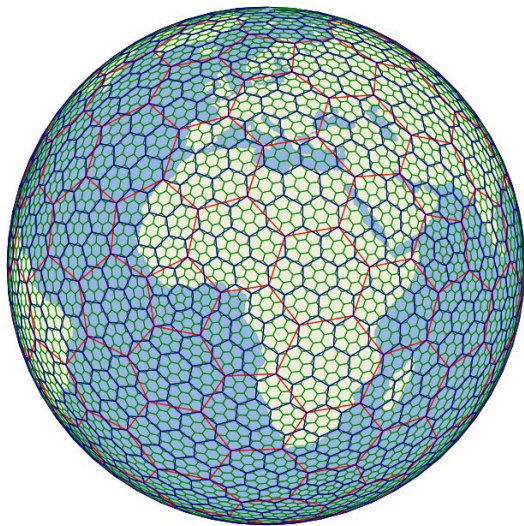


ISEA4H

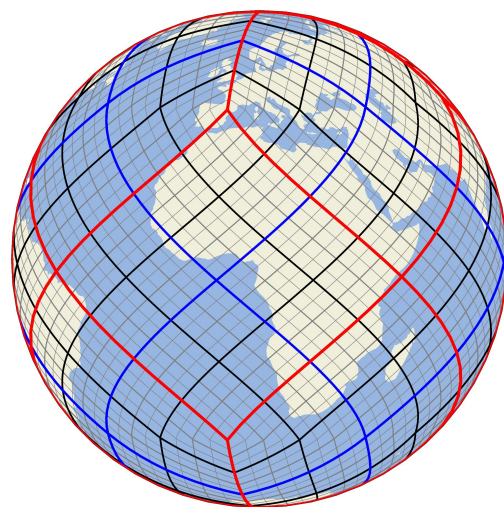
Examples for DGGS



Google S2



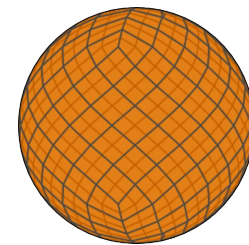
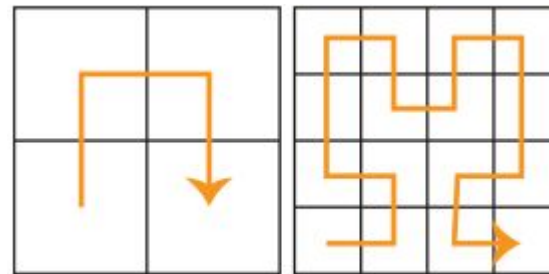
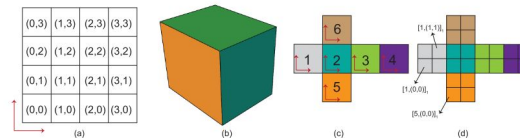
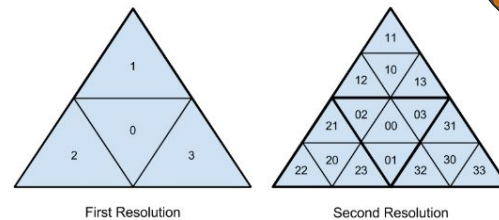
Uber H3



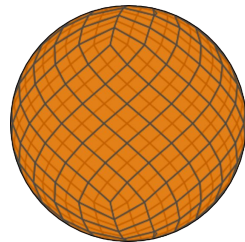
HEALPix

Indexing scheme

- Allows assigning a numerical index to each cell
- Different choices (top to bottom):
 - Hierarchical indexing
 - Axis-based indexing (“Coordinates”)
 - Space-filling curve
- Position on the grid: a single 64-bit integer
- Allows for lots of operations:
 - Hierarchical access (parents, children, siblings)
 - Neighbours search
 - Alignment

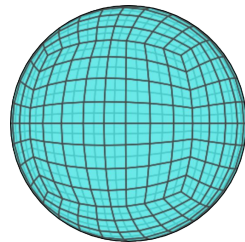


Summary



- DGGS recursively subdivide the globe into smaller parts
- Cells can be arranged into a hierarchy
- Unique IDs allow exact alignment and neighbourhood operations

xdggs



Common API for using DGGs with xarray

Features:

- Metadata encoding scheme (creates a **pandas** index)
- Derivation of cell centers / cell boundaries from cell ids
- Selection based on cell centers
- Built-in support for HEALPix (via **cdshealpix**) and H3 (via **h3ronpy**), more via extension packages (e.g. **xdggs-dggrid4py**)
- Exploratory visualization using **lonboard**

Planned:

- Selection based on parent cell ids or geometries
- Computation of parent / child cell ids / neighbours
- Coarsening / refining
- Memory efficient indexes
- Distributed variants

 [xarray-contrib/xdggs](https://github.com/xarray-contrib/xdggs)

PyPI: xdggs

conda-forge: xdggs

Discord: `discord.gg/M4WPSwDF6S`

Thank you!